Notes: Gibbons & Murphy (JPE, 1992)

Optimal Incentive Contracts in the Presence of Career Concerns: Theory and Evidence

Contents

1 Big picture

- **Question.** How should explicit incentive contracts be designed when workers already face implicit incentives from career concerns? More concretely, how should CEO pay-performance sensitivity vary over a CEO's career when current performance affects the market's belief about ability and thus future compensation?
- **Core mechanism.** Current output is informative about unobserved ability. A worker therefore has an incentive to exert effort today to improve the market's posterior belief about ability. This reputational motive is strongest when the worker has many periods of future compensation at stake.
- **Main theoretical result.** Optimal contracts optimize *total incentives*: explicit contractual incentives plus implicit career-concern incentives. Since implicit incentives are strongest early in a career and weakest close to retirement, explicit pay-performance incentives should be weakest early and strongest near retirement.
- **Key empirical prediction.** CEOs near retirement should have a stronger pay-performance relation than CEOs with more years left in office.
- **Empirical setting.** The paper uses longitudinal CEO compensation data from *Forbes* for 1970–1988, matched to firm performance data from Compustat.
- **Main empirical result.** In elasticity specifications, the interaction between shareholder return and a dummy for CEOs in their last three years is positive and statistically significant. The estimated pay-performance elasticity is higher for CEOs close to retirement.
- **Why this matters.** The paper reconciles the career-concerns view of Fama and Holmstrom with explicit contracting: labor-market discipline and incentive contracts are complements in a dynamic incentive system, not substitutes.
- **Broader lesson.** Compensation contracts should account for all incentive sources. A low explicit incentive slope need not mean weak total incentives if promotion or reputation incentives are strong.

2 Incremental contribution of the paper

- **Adds explicit contracts to career-concerns models.** Earlier career-concerns models emphasized labor-market discipline and reputation. This paper asks how formal pay-for-performance contracts should be designed when reputation incentives are already present.
- **Introduces the total-incentives principle.** The central insight is that the optimal contract does not choose explicit incentives in isolation. It chooses explicit incentives to complement and partly offset implicit incentives created by learning about ability.
- **Links dynamic agency theory to CEO compensation.** Instead of focusing only on the average pay-performance sensitivity, the paper predicts cross-sectional variation: incentive pay should vary systematically with horizon to retirement.
- **Moves the empirical question from "how strong is pay-performance pay?" to "where should it be strongest?"** This is important because the average level of CEO pay-performance sensitivity had already been studied; the new contribution is using career horizon as a test of the theory.
- **Clarifies the role of risk aversion.** Without risk aversion or other costs of incentive provision, explicit contracts could fully neutralize career-concern distortions. The model keeps risk sharing in the problem, so implicit career incentives remain relevant.
- **Builds a bridge between external and internal labor markets.** Although the formal model is external-market based, the interpretation applies to promotions and internal labor markets: workers with more promotion opportunities already have stronger implicit incentives.

- **Provides early evidence on dynamic incentive design.** The empirical strategy tests whether contract slopes change as CEOs approach retirement, which is a direct dynamic implication rather than a static agency prediction.
- **Contributes to the executive-compensation literature.** It shows that CEO contracts cannot be interpreted properly without accounting for career stage, tenure, and the market's uncertainty about managerial ability.

3 Model objects and measurement

3.1 Output, ability, effort, and noise

In each period t, the worker produces output

$$y_t = \eta + a_t + \varepsilon_t, \tag{1}$$

where η is ability, a_t is nonnegative effort, and ε_t is noise. Ability is initially unknown but normally distributed. Output is observed, so the market uses output to update beliefs about ability.

Interpretation: The model makes output a noisy signal of ability and effort. Since effort is not directly observable, the market cannot perfectly separate "the worker is talented" from "the worker worked hard." That creates career concerns.

3.2 Preferences and effort cost

The worker has exponential utility over discounted wage net of effort cost:

$$U(w_1, \ldots, w_T; a_1, \ldots, a_T) = -\exp\left[-r \sum_{t=1}^{T} \delta^{t-1}\{w_t - g(a_t)\}\right]. \tag{2}$$

The cost function $g(a)$ is convex, with $g'(0) = 0$ and increasing marginal cost.

Interpretation: Exponential utility and normal shocks make the model tractable. Risk aversion matters because incentive contracts impose risk on the worker, so the optimal slope trades off incentives against insurance.

3.3 Linear short-term contracts

Each period contract is linear in output:

$$w_t(y_t) = c_t + b_t y_t. \tag{3}$$

The slope b_t is the explicit incentive intensity. Given a linear contract, the worker's direct contractual incentive satisfies

$$g'(a_t) = b_t \tag{4}$$

when no future reputational effect is present.

Interpretation: The slope b_t is the empirical object measured through pay-performance sensitivity or elasticity. The theory predicts how b_t changes with career horizon.

3.4 Belief updating and career concerns

After observing output, the market updates its belief about ability. If the worker exerts more effort today, output rises, the market's posterior mean of ability rises, and future wages rise. The worker therefore receives an implicit incentive even when current pay is not explicitly tied to output.

Interpretation: Career concerns are strongest when future wages matter most and when ability is still uncertain. They weaken as the worker approaches retirement and as the market learns more about ability.

3.5 Empirical measurement of CEO horizon

The empirical analysis measures career horizon using years remaining as CEO. The main dummy is *few years left*, equal to one when the CEO is in the last three full fiscal years in office.

Interpretation: The key empirical proxy is imperfect. Leaving office may mean retirement, death, forced turnover, or transition to board service. The authors therefore document that most departing CEOs appear to end active CEO careers rather than move to another CEO job.

4 Models and theoretical specifications

4.1 Static benchmark: optimal explicit incentive without career concerns

In the final period there is no future labor-market reward from manipulating the market's belief. The optimal incentive slope b^* solves a standard risk-incentive tradeoff:

$$b^* = \frac{1}{1 + r\sigma^2 g''(a^*(b^*))}, \tag{5}$$

where σ^2 represents output risk and r is risk aversion.

Interpretation: A higher slope improves effort incentives but exposes the worker to more output risk. More risk aversion or more output noise lowers the optimal explicit incentive slope.

4.2 Two-period career-concerns mechanism

In a two-period version, first-period effort affects both current pay and the market's posterior belief, which affects the second-period wage. The total first-period incentive is

$$B_1 = b_1 + \delta(1 - b_2^*) \times \text{learning weight}, \tag{6}$$

where b_1 is the explicit first-period slope and the second term is the implicit career-concern incentive.

Interpretation: The career-concern term is positive because higher first-period output raises future expected wages. It is multiplied by $(1 - b_2^*)$ because only the intercept part of future pay reflects the market's updated belief when second-period explicit incentives are optimized.

4.3 Optimal contract with career concerns

The optimal contract chooses b_1 taking into account that total incentives are already partly provided by the career-concern term. Thus the contract solves for an optimal total incentive B_1^* rather than an optimal explicit slope alone:

$$b_1^* = B_1^* - \text{implicit career-concern incentive}. \tag{7}$$

Interpretation: When implicit incentives are strong, the optimal explicit incentive is low. For young workers, current pay may optimally be nearly independent of current performance. For older workers near retirement, career-concern incentives are weak, so explicit pay-for-performance should be stronger.

4.4 Multi-period result

In the T-period model, the same logic generalizes. Implicit incentives decline as the worker approaches the end of the career. The optimal explicit contract slopes therefore increase as retirement approaches:

$$b_1^* < b_2^* < \cdots < b_T^*. \tag{8}$$

Interpretation: This monotonicity generates the empirical prediction that CEOs near retirement should have more sensitive pay-performance contracts.

4.5 Renegotiation-proof long-term contracts

The appendix shows that the sequence of one-period contracts delivers the same incentives as optimal renegotiation-proof long-term contracts in the two-period case. The second-period slope must be the static optimal slope, and the first-period slope again implements the same total incentive.

Interpretation: The central result is not merely an artifact of assuming only short-term contracts. Even when long-term contracts are feasible but renegotiation-proof, optimal incentives still account for career concerns.

5 Data and empirical specifications

5.1 CEO sample

The data come from *Forbes* executive compensation surveys from 1971 to 1989, covering fiscal years 1970–1988. The broad sample includes 2,972 executives serving in 1,493 large corporations. The main completed-spells sample restricts to CEOs who leave office during the sample period and have sufficient compensation and performance data.

Interpretation: Completed spells allow the authors to know how many years a CEO had left in office. This is crucial for testing the retirement-horizon prediction.

5.2 Compensation and performance

The main compensation measure is salary plus bonus in 1988 dollars. Performance is measured by shareholder wealth changes or shareholder returns. Let V_{it} be firm value and r_{it} shareholder return. The paper defines

$$\Delta V_{it} = V_{i,t-1} r_{it}, \qquad \Delta \ln(V_{it}) = \ln(1 + r_{it}). \tag{9}$$

Interpretation: The dollar specification is sensitive to firm size because large firms have larger dollar wealth changes. The elasticity specification uses percentage performance and helps control for size-related heterogeneity.

5.3 Empirical specification from the model

The model implies that changes in compensation depend on current and lagged performance. The empirical specification is summarized as

$$\Delta w_{it} = \alpha_t + \beta_t \Delta V_{it} + \gamma_t \Delta V_{i,t-1} + u_{it}, \tag{10}$$

or in elasticity form:

$$\Delta \ln(w_{it}) = \alpha + \text{year effects} + \beta \Delta \ln(V_{it}) + u_{it}. \tag{11}$$

Interpretation: The lagged-performance term comes from belief updating: last year's performance affected the market's belief and current wage intercept. In practice, the key prediction is about how the current pay-performance coefficient varies with years remaining.

5.4 Main testing equation

The main regression can be written as

$$\Delta \ln(w_{it}) = \alpha + \alpha_1 FewYearsLeft_{it} + \text{year dummies} + [\beta_0 + \beta_1 FewYearsLeft_{it} + \text{year interactions}] \Delta \ln(V_{it}) + u_{it}. \tag{12}$$

The coefficient of interest is β_1.

Interpretation: A positive β_1 means that CEOs close to retirement have a stronger pay-performance relation, as predicted by the career-concerns model.

6 Identification and empirical design

6.1 Core identifying comparison

The empirical design compares CEO-year observations close to retirement with CEO-year observations farther from retirement. The identifying variation is within the distribution of CEO career horizons and firm performance shocks.

Interpretation: The test is not a randomized experiment. It is a theory-guided cross-sectional and panel comparison of pay-performance slopes across career horizons.

6.2 Why retirement horizon is informative

The model predicts that career concerns depend on the number of future periods over which reputation can affect wages. CEOs in their last three years have less future labor-market value at stake, so implicit incentives are weaker.

Interpretation: The same observed stock return should be associated with a larger current pay response when the CEO is near retirement if explicit contracts compensate for weaker implicit incentives.

6.3 Main threats

- **Firm size.** Dollar performance changes are larger in large firms, potentially biasing pay-performance sensitivities.

- **CEO selection.** CEOs near retirement may differ systematically in ability, tenure, or firm assignment.
- **Unobserved firm heterogeneity.** Firms may have different compensation philosophies or governance systems.
- **Completed-spells selection.** Knowing years remaining requires observing departure, so the main sample is not all CEOs.
- **Retirement versus turnover.** Departure may reflect forced turnover or health rather than planned retirement.

6.4 How the paper addresses threats

The paper uses logarithmic pay and return specifications to reduce firm-size bias, includes year dummies and year-by-performance interactions, checks low-tenure effects, compares completed spells with the full sample, and examines specifications that allow firm-specific pay-performance slopes.

Interpretation: These robustness checks do not eliminate all identification concerns, but they show that the main positive near-retirement interaction is not simply a mechanical artifact of firm size or sample construction.

7 Section-by-section study guide

- **Introduction.** Explains the total-incentives idea: explicit contracts and career concerns jointly determine effort incentives.
- **Theoretical analysis.** Builds a dynamic agency model with ability learning, risk aversion, and linear contracts. The main result is increasing explicit incentives over the career.
- **Evidence on CEOs.** Constructs CEO data, documents career horizons, and estimates pay-performance relations by years remaining in office.
- **Conclusion and appendix.** Discusses robustness of the theoretical result and shows that renegotiation-proof long-term contracts implement the same incentives.

8 Tables and figures

8.1 Figure 1: Frequency distribution for CEO tenure, firm tenure, age, and years remaining in office

Read:

- Panel A shows the distribution of years as CEO when leaving office. Mean tenure as CEO is about 9.8 years, with median 7.9 years.
- Panel B shows years in the firm when leaving office. CEOs typically have long firm tenure, with mean about 28.1 years and median about 30.9 years.
- Panel C shows age at CEO departure. Many CEOs leave near ages 64–65, consistent with retirement rather than lateral movement to another CEO job.
- Panel D shows years remaining as CEO for CEO-year observations in the completed-spells sample. Many observations are close to the end of the spell, which provides variation for the empirical test.

Detailed interpretation: The figure validates the empirical proxy for career horizon. If CEOs often left to take comparable jobs elsewhere, "years remaining as CEO" would not measure retirement horizon. Instead, the distribution shows long firm attachment and departure around normal retirement ages, supporting the interpretation that later years in the sample are years with weak future career concerns.

Economic message: The career-concerns prediction requires that CEOs near the end of their CEO spell have fewer future reputational rents at stake. Figure 1 supports this assumption and motivates the few-years-left dummy.

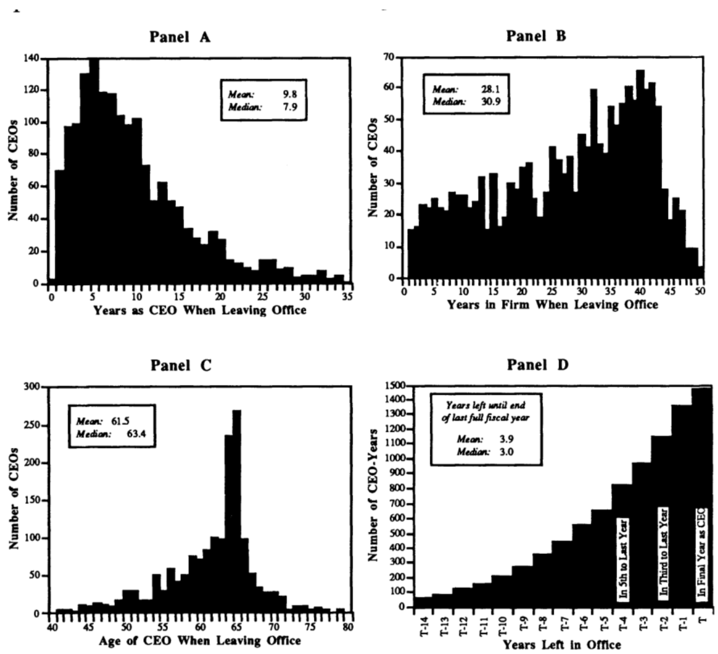

FIG. 1.—Frequency distribution for CEO tenure, firm tenure, age, and years remaining in office for CEOs leaving office during the 1970–88 sample period. Histograms and statistics are based on 1,631 CEOs, representing 916 firms and 8,786 CEO-years, who leave their firms during the 1970–88 sample period. Panel *A* does not include 23 CEOs who left after more than 34 years as CEO; panel *B* does not include 18 CEOs who left after more than 50 years in their firm; panel *C* does not include eight CEOs who left after age 80; and panel *D* does not include 107 CEO-years with more than 14 years remaining (this panel is based only on those who actually leave office during the sample period; these CEOs have *at most* 18 years remaining).

Figure 1: Frequency distribution for CEO tenure, firm tenure, age, and years remaining in office.

8.2 Table 1: Tenure, Compensation, and Corporate Summary Statistics for the Completed-Spells Sample

Read:

- The completed-spells sample has 6,737 CEO-year observations: 3,117 in the last three years and 3,620 not in the last three years.
- CEOs in their last three years are older, have longer firm tenure, and have longer CEO tenure than CEOs farther from retirement.
- Near-retirement CEOs have higher average salary plus bonus: about \$592,800 versus \$536,400 for other completed-spells CEOs.
- The standard deviation of compensation changes is higher for near-retirement CEOs, consistent with stronger pay-performance sensitivity.

- Firm characteristics are similar across the two groups in terms of sales, market value, returns, and shareholder wealth changes, though there are some differences in fiscal years.

Detailed interpretation: Table 1 establishes the baseline sample and shows that the two key groups differ in career horizon but are not obviously incomparable in firm performance. The table also foreshadows the main result: compensation variability is higher near retirement, which is consistent with stronger incentive pay.

Economic message: The table supports the paper's identifying comparison. It shows a meaningful near-retirement group and a comparison group with similar firm performance, allowing the pay-performance slope test to focus on incentive design rather than raw performance differences.

TABLE 1

TENURE, COMPENSATION, AND CORPORATE SUMMARY STATISTICS FOR "COMPLETED-SPELLS SAMPLE" OF CEOs LEAVING OFFICE DURING THE 1970–88 SAMPLE PERIOD, GROUPED BY YEARS REMAINING AS CEO

Variable	All CEOs Leaving Office during Sample Period (1)	CEOs in Their Last Three Years in Office (2)	CEOs Not in Their Last Three Years in Office (3)	Test Statistic for Difference* (4)
Number of CEO-years	6,737	3,117	3,620	
	A. CEO Characteristics			
Mean age	59.1	61.2	57.3	t = 29.4*
Mean tenure in firm	26.8	28.4	25.4	t = 10.8*
Mean tenure as CEO	8.9	9.6	8.2	t = 8.6*
Mean years remaining as CEO until end of final fiscal year†	3.8	1.0	6.2	t = 93.7*
	B. Compensation Statistics			
Mean salary + bonus	$562,500	$592,800	$536,400	t = 8.0*
Mean total compensation‡	$616,300	$649,000	$588,200	t = 6.8*
Δ(salary + bonus): Mean	$25,500	$28,400	$22,900	t = 1.8
Standard deviation	$123,300	$140,400	$106,500	F = 1.7*
Δ%(salary + bonus): Mean	6.6%	7.0%	6.2%	t = 1.3
Standard deviation	23.4%	26.7%	20.0%	F = 1.8*

Table 1: Completed-spells sample summary statistics, panels A and B.

	C. Firm Characteristics (Millions of Dollars)			
Mean total sales	$4,239	$4,417	$4,087	t = 1.5
Mean market value of stock	$2,352	$2,511	$2,216	t = 2.1*
Δ(shareholder wealth): Mean	−$53	−$19	−$83	t = 1.4
Standard deviation	$1,892	$2,077	$1,717	F = 1.5*
Shareholder return:§ Mean	1.8%	1.8%	1.8%	t = .1
Standard deviation	31.5%	31.2%	31.8%	F = 1.0*
Mean fiscal year	1977.7	1979.1	1976.5	t = 24.4

NOTE.—The sample is constructed from longitudinal data reported in *Forbes* on 1,292 CEOs serving in 785 firms who leave their firms during the 1970–88 sample period. Monetary variables are in 1988-constant dollars.

* The difference between cols. 2 and 3 is significant at the 5 percent level. F-statistics test the equality of subsample variances with 3,116 and 3,619 degrees of freedom.

† Tenure as CEO and years remaining as CEO are evaluated in the middle of the fiscal year. Thus a CEO retiring at fiscal year end would have one-half a year remaining. This measure understates actual years remaining since it ignores time served after the end of the CEO's final full fiscal year.

‡ Total compensation includes salary, bonus, value of restricted stock, savings and thrift plans, and other benefits but does *not* include the value of stock options granted or the gains from exercising stock options.

§ Cumulative rate of return on common stock, defined as log(1 + annual return).

Table 1: Completed-spells sample summary statistics, panel C and notes.

8.3 Figure 2: Estimated pay-performance elasticities, by years remaining as CEO

Read:

- The bars plot estimated pay-performance elasticities for CEOs grouped by years remaining in office.
- The final three years have high elasticities: roughly 0.178 in the final year, 0.203 in the second-to-last year, and 0.183 in the third-to-last year.
- The fifth- and sixth-to-last years have lower elasticities, about 0.119 and 0.110.
- The figure visually supports the model prediction that explicit pay-performance incentives rise near retirement.
- Standard errors are shown in the bars; the near-retirement bars are high relative to many middle-career categories.

Detailed interpretation: Figure 2 is the visual centerpiece of the empirical evidence. It translates the dynamic-contract prediction into a simple pattern: explicit incentives should be strongest where career concerns are weakest. The cluster of high elasticities in the last three years is consistent with that prediction.

Economic message: The figure is not about whether CEO pay is generally sensitive enough to performance. It is about how incentive intensity changes over the CEO life cycle. That is the incremental empirical insight of the paper.

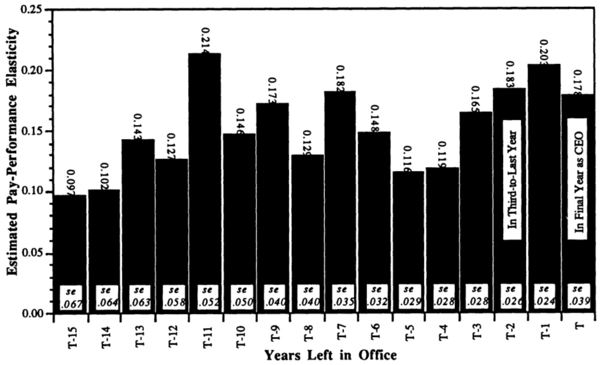

FIG. 2.—Estimated pay-performance elasticities, by years remaining as CEO. The figure depicts estimated coefficients from eq. (34) and is drawn under the assumption of a year effect of .06, which is the average of the 17 estimated year coefficients. Standard errors for each estimated coefficient are in italics. The group $T - 15$ includes a small number of CEOs with more than 15 years remaining.

Figure 2: Estimated pay-performance elasticities, by years remaining as CEO.

TABLE 2

COEFFICIENTS OF ORDINARY LEAST SQUARES REGRESSIONS OF Δ ln(CEO SALARY + BONUS) ON SHAREHOLDER RETURN, INCLUDING INTERACTIONS TO ALLOW THE PAY-PERFORMANCE RELATION TO VARY WITH YEAR, YEARS AS CEO, AND YEARS REMAINING AS CEO

		Dependent Variable: Δ ln(CEO Salary + Bonus)		
		Subsample of CEOs Who Retire 1975–88		Full Sample
Independent Variable	Predicted Sign	(1)	(2)	(3)
Few years left* (dummy variable)		−.0090	−.0056	−.0085
		(−2.0)	(−1.2)	(−2.0)
Low tenure† (dummy variable)		...	.0396	.0435
			(8.4)	(11.2)
Can't tell if few years left‡ (dummy variable)		...	...	−.0095
				(−1.4)
Few years left × shareholder return	+	.0436	.0429	.0383
		(3.0)	(3.0)	(2.9)
Low tenure × shareholder return	−	...	.0032	.0199
			(.2)	(1.6)
Can't tell if few years left × shareholder return		...	...	.0335
				(1.8)
R^2		.0833	.0929	.0927
Sample size		6,730	6,730	11,360

NOTE.—The sample is constructed from longitudinal data reported in *Forbes* on 2,236 CEOs serving in 1,204 firms during 1970–88. The subsample in col. 1 includes 1,292 CEOs serving in 785 firms leaving their firms during the sample period. Compensation is measured in thousands of 1988-constant dollars. All regressions also include intercepts, year dummies, shareholder return and shareholder return interacted with the year dummies, allowing the pay-performance elasticity to vary by year. t-statistics are in parentheses.

* Few years left is a dummy variable that equals one if the CEO is in his last three years as CEO.

† Low tenure is a dummy variable that equals one if the CEO is in his first four years as CEO.

‡ Can't tell if few years left is a dummy variable that equals one if the CEO-year observation is in the last three years of the data on a CEO who does *not* belong to the completed-spells subsample.

Table 2: OLS regressions of $\Delta \ln$(CEO salary + bonus) on shareholder return and career-horizon interactions.

8.4 Table 2: OLS regressions of $\Delta \ln$(CEO Salary + Bonus) on shareholder return with career-horizon interactions

Read:

- Column 1 uses the completed-spells subsample and estimates a positive coefficient of 0.0436 on shareholder return × few years left, with a t-statistic of 3.0.
- Column 2 adds a low-tenure dummy and a low-tenure interaction. The few-years-left interaction remains positive at 0.0429 with a t-statistic of 3.0.
- The low-tenure interaction has the wrong sign for the second hypothesis and is insignificant in the completed-spells sample.
- Column 3 uses the full sample, including CEOs whose final date is not observed. The few-years-left interaction remains positive at 0.0383 with a t-statistic of 2.9.
- The few-years-left dummy itself is negative, meaning average pay growth is lower near retirement, but the pay-performance slope is higher.

Detailed interpretation: Table 2 provides the main regression evidence for Hypothesis 1. The same shareholder return produces a larger pay change when the CEO is close to retirement. The robustness across completed-spells and full-sample specifications is important because the completed-spells sample could otherwise be viewed as selected.

Economic message: The table supports the total-incentives view: when implicit career incentives are low, explicit incentive contracts become more performance-sensitive. The weak evidence for Hypothesis 2 suggests that the cleanest empirical support is for retirement horizon rather than for tenure-based learning about ability.

8.5 Table 3: Regressions allowing the pay-performance relation to vary with firm size

Read:

- Columns 1–2 use dollar compensation changes and dollar shareholder wealth changes. The few-years-left interaction is positive but not significant in dollar terms.
- Adding interactions with sales and sales squared changes the dollar-pay coefficient, showing that size-related heterogeneity matters in dollar specifications.
- Columns 3–4 use log changes. The shareholder return × few years left coefficient is 0.0436 in column 3 and 0.0432 in column 4, both with t-statistics of 3.0.
- The size interactions in the log specification do not materially change the career-horizon coefficient.

Detailed interpretation: Table 3 addresses a major concern: pay-performance sensitivity in dollars naturally differs by firm size. The log specification controls for this problem. The fact that the near-retirement interaction remains stable after allowing firm-size interactions strengthens the interpretation of the main result.

Economic message: Career-concern effects are most cleanly detected in elasticity form. This table shows that the main result is not driven by CEOs near retirement being concentrated in firms of a particular size.

TABLE 3

COEFFICIENTS OF ORDINARY LEAST SQUARES REGRESSIONS OF Δ(CEO Salary + Bonus) ON Δ(Shareholder Wealth) AND Δ ln(CEO Salary + Bonus) ON SHAREHOLDER RETURN, INCLUDING INTERACTIONS TO ALLOW THE PAY-PERFORMANCE RELATION TO VARY WITH YEARS REMAINING AS CEO AND WITH FIRM SIZE AS MEASURED BY SALES

	Dependent Variable: Δ(CEO Salary + Bonus)		Dependent Variable: Δ ln(CEO Salary + Bonus)	
Independent Variable	(1)	(2)	(3)	(4)
Few years left* (dummy variable)	−3.06	−2.63	−.0090	−.0091
	(−1.0)	(−.9)	(−2.0)	(−2.0)
Δshareholder wealth × sales	. . .	-4.0×10^{-7}	. . .	. . .
		(−4.2)		
Δshareholder wealth × (sales)^2	. . .	3.1×10^{-12}	. . .	. . .
		(3.7)		
Δshareholder wealth × few years left	.0008	.0025	. . .	. . .
	(.4)	(1.2)		
Shareholder return × sales	. . .	. . .	. . .	2.7×10^{-6}
				(1.5)
Shareholder return × (sales)^2	. . .	. . .	. . .	-3.2×10^{-12}
				(−.1)
Shareholder return × few years left	. . .	. . .	.0436	.0432
			(3.0)	(3.0)
R^2	.0426	.0452	.0833	.0845
Sample size	6,727	6,727	6,730	6,730

NOTE.—The sample is constructed from longitudinal data reported in *Forbes* on 1,292 CEOs serving in 785 firms who leave their firms during the 1970–88 sample period. Compensation is measured in thousands of 1988-constant dollars; Δ(shareholder wealth) and sales are measured in millions of 1988-constant dollars. All regressions include intercepts and year dummies. Cols. 1 and 2 include Δshareholder wealth and Δshareholder wealth interacted with the year dummies, allowing the pay-performance sensitivity to vary by year; cols. 3 and 4 include shareholder return and shareholder return interacted with the year dummies, allowing the pay-performance elasticity to vary by year. t-statistics are in parentheses.

* Few years left is a dummy variable that equals one if the CEO is in his last three years as CEO.

Table 3: Pay-performance relation with firm-size interactions.

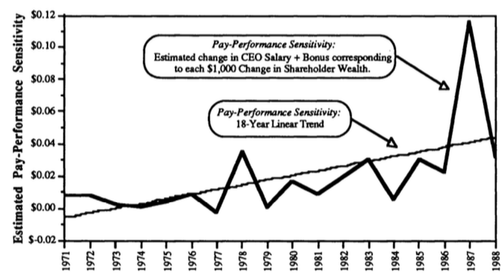

FIG. A1.—Estimated pay-performance sensitivities, by year. Figure is based on annual regressions of Δ(salary + bonus [$ thousands]) on Δ(shareholder wealth [$ millions]) using *Forbes* data on 2,236 CEOs serving in 1,204 firms. Each year's regression is based on approximately 650 observations.

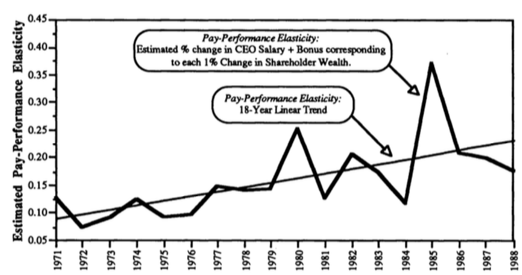

FIG. A2.—Estimated pay-performance elasticities, by year. Figure is based on annual regressions of Δ ln(salary + bonus [$ thousands]) on Δ ln(shareholder wealth [$ millions]) using *Forbes* data on 2,236 CEOs serving in 1,204 firms. Each year's regression is based on approximately 650 observations.

Appendix Figures A1–A2: Estimated pay-performance sensitivities and elasticities by year.

8.6 Appendix Figures A1–A2: Estimated pay-performance relations by year

Read:

- Figure A1 plots annual pay-performance sensitivities in dollar terms.
- Figure A2 plots annual pay-performance elasticities.
- Both figures show that pay-performance relations trend upward over the sample period.
- These trends justify year effects and year-by-performance interactions in the empirical specifications.

Detailed interpretation: The appendix figures show why the regressions must control for secular changes in compensation practices. If near-retirement CEOs are disproportionately observed in later years, time trends in incentive pay could masquerade as career-concern effects. The year-interaction controls help address this issue.

Economic message: The figures indicate that executive incentive pay became stronger during the sample period. The paper's main evidence is therefore not just a time-series fact; it survives controls for year-specific pay-performance relations.

9 Result map

- **Result 1: Career concerns create implicit incentives.** Because output affects future beliefs about ability, effort has future wage consequences.
- **Result 2: Explicit and implicit incentives are substitutes in the optimal contract.** The contract slope is lower when career concerns are stronger.
- **Result 3: Explicit incentives rise as retirement approaches.** Since future reputational benefits shrink near retirement, the optimal explicit pay-performance slope increases.
- **Result 4: CEO data support the retirement-horizon prediction.** Pay-performance elasticities are higher for CEOs in their last three years.
- **Result 5: Firm-size controls matter, but do not overturn the main elasticity result.** Dollar pay-performance sensitivities are size-sensitive; log elasticities are more robust.
- **Result 6: Tenure-based learning evidence is weaker.** The low-tenure interaction does not strongly support the hypothesis that incentives rise with tenure holding years remaining fixed.

10 Short summary

- The paper studies optimal incentive contracts when workers have career concerns.
- Career concerns arise because output updates the market's belief about ability.
- A worker far from retirement has strong implicit incentives because current output affects many future wages.
- A worker close to retirement has weak implicit incentives because fewer future wages remain.
- The optimal contract optimizes total incentives: explicit contractual pay-performance incentives plus implicit reputation incentives.
- The model predicts that explicit incentives should be strongest near retirement.
- The empirical analysis uses CEO compensation and firm performance data from 1970–1988.
- The main proxy for horizon is whether the CEO is in the last three years in office.
- CEO pay-performance elasticities are higher in the last three years.

- The result survives controls for year effects and firm-size heterogeneity in log specifications.
- The paper's broader contribution is to show that contracts should be interpreted in the context of other incentive systems, especially career concerns and labor-market learning.

11 Conclusion

11.1 Core conclusion

Gibbons and Murphy show that career concerns remain important even when explicit compensation contracts are available. The optimal contract does not simply solve a static moral-hazard problem; it adjusts explicit incentives to complement implicit incentives generated by labor-market learning. The main qualitative implication is simple and powerful: explicit performance pay should be strongest when career concerns are weakest.

11.2 Economic intuition

Early in a career, current output can change the market's belief about ability and thereby affect many future wages. This gives the worker strong implicit incentives even if current pay is not very sensitive to current performance. Near retirement, the same output signal has much less future value. The worker therefore needs stronger explicit incentives to achieve the same total incentive intensity.

11.3 Contribution to incentive theory

The paper integrates five elements that were often studied separately: incentives, learning, market forces, contracts, and risk aversion. This integration matters because career concerns without contracts can generate too much early effort and too little late effort, while contracts without career concerns miss the implicit incentive environment. The contribution is the total-incentives principle: the explicit contract should be designed after accounting for implicit market incentives.

11.4 Contribution to executive compensation

The paper shifts attention from the average magnitude of CEO pay-performance sensitivity to its cross-sectional variation. The prediction is not merely that CEOs should be paid for performance, but that CEOs near retirement should be paid more strongly for performance because their career-concern incentives are weaker. This is a sharper and more theory-specific test.

11.5 Limitations

The CEO application is not a perfect laboratory. CEOs often have long histories inside their firms before appointment, so ability may already be partly known. CEO labor markets are not perfectly competitive, and CEOs rarely move directly from one CEO job to another. Departure dates may reflect forced turnover as well as retirement. The evidence is therefore suggestive rather than decisive.

11.6 Takeaway

The lasting lesson is that incentive contracts cannot be evaluated in isolation. Observed pay-performance sensitivity is only one part of a broader incentive system. Promotion opportunities, labor-market reputation, ability learning, and career horizon all affect how much explicit performance pay is optimal.